

ALPACA AI TRADING AGENTS HACKATHON · 2026

# Capitol Desk

Faaris Ahmed

An autonomous options desk that reads congressional financial disclosures, decides whether the signal is still alive, and trades it — end to end, unattended.

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[capitol-desk.onrender.com](https://capitol-desk.onrender.com) · Claude Opus 5 · Alpaca MCP Server · Alpaca Trading API

# Congress trades. The law says they must tell us.

The STOCK Act requires every member of Congress to publicly disclose any transaction over \$1,000 — within 45 days. Those filings are free, public, and almost unreadable.

**368**

Periodic Transaction Reports filed in 2026

**31.5** days

Median lag from trade to disclosure

**12%**

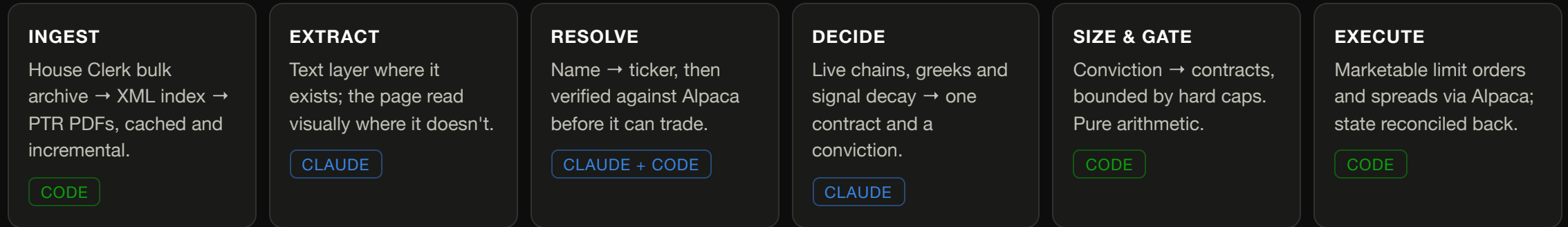
Filed as scanned paper — zero machine-readable text

Public filings. Trading on them is not insider trading — it is the opposite: acting on information the law requires be published.

- Our answer:** Claude reads the text layer where one exists, and reads the **page visually** where one doesn't. On that 274-row scan it extracted every transaction — and correctly ignored the example row.

(This page will be publicly disclosed )

# Six stages, and the model only touches two



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**Claude decides *what* to trade. Code decides *how much*.**

Position sizing and risk limits are plain arithmetic in a config file and a gate function. There is no prompt a model can write that gets it a bigger position.

# A real filing, a real order



## PERIODIC TRANSACTION REPORT

Filing ID #20035143

Clerk of the House of Representatives • Legislative Resource Center • B81 Cannon Building • Washington, DC 20515

### FILER INFORMATION

**Name:** Hon. Nancy Pelosi  
**Status:** Member  
**State/District:** CA11

### TRANSACTIONS

| ID | Owner | Asset                                                                                                                         | Transaction Type | Date       | Notification Date | Amount                    | Cap. Gains > \$200?      |
|----|-------|-------------------------------------------------------------------------------------------------------------------------------|------------------|------------|-------------------|---------------------------|--------------------------|
| SP |       | Bloom Energy Corporation Class A Common Stock (BE) [ST]                                                                       | P                | 07/24/2026 | 07/24/2026        | \$1,000,001 - \$5,000,000 | <input type="checkbox"/> |
|    |       | FILING STATUS: New<br>DESCRIPTION: Purchased 10,000 shares.                                                                   |                  |            |                   |                           |                          |
| SP |       | Bloom Energy Corporation Class A Common Stock (BE) [OF]                                                                       | P                | 07/24/2026 | 07/24/2026        | \$1,000,001 - \$5,000,000 | <input type="checkbox"/> |
|    |       | FILING STATUS: New<br>DESCRIPTION: Purchased 100 call options with a strike price of \$100 and an expiration date of 6/17/27. |                  |            |                   |                           |                          |
| SP |       | Bloom Energy Corporation Class A Common Stock (BE) [ST]                                                                       | P                | 07/28/2026 | 07/28/2026        | \$500,001 - \$1,000,000   | <input type="checkbox"/> |
|    |       | FILING STATUS: New<br>DESCRIPTION: Purchased 5,000 shares.                                                                    |                  |            |                   |                           |                          |
| SP |       | Bloom Energy Corporation Class A Common Stock (BE) [ST]                                                                       | P                | 07/28/2026 | 07/28/2026        | \$500,001 - \$1,000,000   | <input type="checkbox"/> |
|    |       | FILING STATUS: New<br>DESCRIPTION: Purchased 5,000 shares.                                                                    |                  |            |                   |                           |                          |

### WHAT THE FILING SAYS

Pelosi, filed 21 Aug. Bloom Energy calls, **\$100 strike, expiring 17 Jun 2027** — a deep in-the-money LEAP used as stock replacement.

### WHAT THE DESK FOUND

BE had already run **+13.9%** since her trade date. The outright June-2027 contract now costs **\$8,116** against a **\$5,000** per-trade cap — and every affordable strike was 100% time value at 88% implied vol: **volatility risk the filer never took.**

### WHAT IT DID

Built a vertical debit spread — long the 175 call (delta 0.75), short the 230. Same expiry, same thesis, **\$2,071 of risk instead of \$8,116**. Conviction cut to 0.45 because the move had partly happened.

# It declines far more than it takes

650

Transactions parsed

635

Passed over

3

Trades placed

## DECLINED ON THE MERITS — VERBATIM

**PG** — "a \$1k–\$15k purchase inside a 401(k) subholding of a family trust: a routine retirement-account allocation into a mega-cap consumer staple, not an informed, concentrated bet ... there's no thesis to express."

**AMAT** — "the quote set is internally incoherent: every listed call is priced above the maximum theoretical value of a call ... there is no way to size or price an honest replication." (It had caught a bug in our own price feed.)

Anyone can build a bot that copies Pelosi. The engineering is in the 635 it refused.

WE MEASURED IT

# Copying Congress indiscriminately does not work

Event study over 143 filings and 826 disclosed purchases. Entry is the **filing** date, not the transaction date — a trade is private for up to 45 days, so entering earlier is lookahead bias. Returns are excess over SPY across the identical window.

| HORIZON         | EVENTS | MEAN EXCESS | MEDIAN EXCESS | HIT RATE |
|-----------------|--------|-------------|---------------|----------|
| 5 trading days  | 799    | +0.09%      | −0.40%        | 45.2%    |
| 21 trading days | 669    | +0.74%      | −0.37%        | 49.0%    |
| 63 trading days | 668    | +0.42%      | +0.08%        | 50.1%    |

Hit rates are a coin flip and the median trade is flat to negative. The positive means come from a handful of large winners, not a broad edge. **That is the argument for the desk, not against it** — if the average disclosed purchase is noise, the value has to come from selection.

# Every limit is code, not a prompt

- **\$5,000** maximum per trade · **\$25,000** per day
- **Never >5%** of a contract's open interest; minimum OI 50
- **Spreads >20% wide are skipped** — a thesis you can't exit is worth nothing
- **Marketable limit orders only.** Never a market order on an option
- **Paper-only guard:** the desk refuses to start against a live endpoint

## CAUGHT IN DEVELOPMENT

Order reconciliation began writing real broker statuses while every limit query still filtered on 'placed'. The daily cap and position limit **silently stopped binding**.

Found, fixed, and pinned with parametrised regression tests asserting that live statuses consume the caps and dead ones release them.

45 tests cover the deterministic machinery — sizing under caps, spread validation, option-symbol parsing, the paper-only guard.



```
pytest tests/ -q
Deterministic machinery under test
[100%]
.....
.venv/lib/python3.12/site-packages/websockets/legacy/__init__.py:6
/Users/faar1s/WORKSPACE/team-shaw-hackathon/.venv/lib/python3.12/site-packages/websockets/legacy/__init__.py:6: DeprecationWarning: websockets.leg
```

The model can argue for a trade. It cannot argue for a bigger one.



RECEIPTS

# Not a mockup

Real filings, real orders, real fills — straight from the terminal.

desk extract 20035143

One filing, fully structured

\$ desk extract 20035143

Hon. Nancy Pelosi CA11 filed 2026-08-21  
7 transactions extracted

P BE [ST] \$1,000,001-\$5,000,000 2026-07-24

P BE [OP] \$1,000,001-\$5,000,000 2026-07-24 -> 100x call \$100 exp 2027-06-17

P BE [ST] \$ 500,001-\$1,000,000 2026-07-28

P BE [OP] \$ 500,001-\$1,000,000 2026-07-28 -> 100x call \$100 exp 2027-06-17

P INTC [OP] \$ 250,001-\$500,000 2026-07-24 -> 50x call \$50 exp 2027-06-17

P INTC [ST] \$ 500,001-\$1,000,000 2026-07-24

P - [AB] \$ 500,001-\$1,000,000 2026-07-27

desk positions

Live book, marked against Alpaca

PA3K1WQNBURG equity \$99,574.87 cash \$92,889.87

| Symbol              | Qty | Avg cost | Current | Market value | P&L       | P&L % |
|---------------------|-----|----------|---------|--------------|-----------|-------|
| BE270617C00175000   | 1   | \$79.55  | \$75.00 | \$7,500.00   | \$-455.00 | -5.7% |
| BE270617C00230000   | -1  | \$59.05  | \$57.10 | \$-5,710.00  | \$195.00  | +3.3% |
| INTC270115C00090000 | 2   | \$14.50  | \$13.95 | \$2,790.00   | \$-110.00 | -3.8% |
| INTC270617C00090000 | 1   | \$21.60  | \$21.05 | \$2,105.00   | \$-55.00  | -2.5% |

RUNNING NOW

# Live, autonomous, and auditable

Positions and P&L stream from a real Alpaca paper account. `desk` `loop` runs unattended — scanning for filings, holding them when the market is closed rather than pricing into a session it hasn't seen, and handing the book to a review agent over **Alpaca's MCP server**.

Capitol Desk

PAPER

PA3K1WQNBWVG

READ-ONLY DEMO

Mirrors public STOCK Act disclosures into options. Updated 09:59:55 EDT.

Positions and account are live from Alpaca. Scanning is disabled here — clone the repo and run `desk` `run` to watch it read filings and trade.

PORTFOLIO VALUE

\$99,614.87

▼ -385.13 today · \$92,890 options buying power

Filings read

44

Transactions parsed

650

Members tracked

36

Trades placed

3

Capital deployed

\$7,154

Passed over

635

OPEN POSITIONS

| Contract            | Qty | Avg cost | Last    | Value       | Unrealised      | P&L |
|---------------------|-----|----------|---------|-------------|-----------------|-----|
| BE270617C00175000   | 1   | \$79.55  | \$76.45 | \$7,645.00  | -310.00 (-3.9%) |     |
| BE270617C00230000   | -1  | \$59.05  | \$59.50 | \$-5,950.00 | -45.00 (-0.8%)  |     |
| INTC270115C00090000 | 2   | \$14.50  | \$14.40 | \$2,880.00  | -20.00 (-0.7%)  |     |
| INTC270617C00000000 | 1   | \$21.60  | \$21.50 | \$2,150.00  | 10.00 (0.5%)    |     |

CAPITOL DESK

# The interesting agent isn't the one that trades. It's the one that declines.

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Live demo

[capitol-desk.onrender.com](https://capitol-desk.onrender.com)

Source

[github.com/faarisaahmed/  
team-shaw-hackathon](https://github.com/faarisaahmed/team-shaw-hackathon)

Built with

Claude Opus 5 · Claude Agent SDK  
Alpaca MCP Server · Alpaca API

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Faaris Ahmed